

Brainstorming & Idea Prioritization

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

As the sole contributor on this project, ideas were listed individually and later grouped by theme before being scored for feasibility and importance.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Venu Gopal	Detect a learner's emotional state (confused, frustrated, bored, confident, curious) directly from the text they type while studying.	Emotion AI	1
2	Venu Gopal	Compare a lightweight BiLSTM classifier against a fine-tuned BERT classifier to see which detects emotion more reliably.	Model Design	1
3	Venu Gopal	Use a keyword-boosting layer to correct obvious misclassifications (e.g. the word 'stuck' should push toward Frustrated).	Model Design	1
4	Venu Gopal	Feed the detected emotion + subject + problem text into Google Gemini to generate a personalised, empathetic learning response.	Generative AI	2
5	Venu Gopal	Provide a rule-based fallback response (emoji + tip + action) if the Gemini API is unavailable, so the app never fails silently.	Reliability	2
6	Venu Gopal	Log every interaction to CSV and show an analytics dashboard (emotion distribution, emotional journey, field-wise trends) to the learner.	Analytics	3

Step 2: Idea Prioritization

Grouped ideas were rated on feasibility (can it realistically be built with the available datasets and time) and importance (how much it improves the learner's experience), then carried forward into the final solution.

Group No.	Final Idea	Feasibility	Importance	Priority Selected
1	Dual-model (BiLSTM + BERT) emotion detection engine with keyword-based confidence boosting	High	High	Yes
2	Gemini-powered personalised learning response with rule-based fallback	High	High	Yes
3	Interaction logging and analytics dashboard (Plotly)	Medium	Medium	Yes